ExoHunter: AI-Powered Exoplanet Candidate Detection and Ranking

Final project report. Data: NASA Exoplanet Archive TESS Object of Interest (TOI) table, downloaded 10 October 2026.

1. Summary

We asked whether machine learning can combine transit characteristics to rank exoplanet candidates better than a conventional signal-detection baseline, and explain each score. On 2,834 candidates already judged real or false, a gradient-boosted model (XGBoost) reached an exam AUC of 0.76 to 0.92 depending on which clues it could see, against 0.55 for a depth-based baseline. SHAP explanations showed that the model also leans on star properties (distance, temperature, size), which reflects selection bias in which candidates get followed up. A distance-matched retraining showed most of the skill survives, but the *top of the ranking* changes a great deal. Real TESS light curves for 30 top-ranked candidates showed a dip in every testable case and no large eclipsing-binary signature, though several of our own early verdicts needed correcting. A pilot adding light-curve measurements to the model gave a small gain whose cause we could not establish.

Bottom line: ML clearly beats the baseline and its scores are explainable, but the scores order candidates only. They are not probabilities of being a planet, and which candidates rank first depends on how selection bias is handled.

2. Problem

When a planet crosses in front of its star, the star's brightness dips slightly, producing a dip in a light curve. A dip does not prove a planet: instrument noise, stellar activity and eclipsing binary stars can mimic transits. Astronomers need to decide which of thousands of candidates deserve follow-up. Our question: can ML combine transit characteristics to rank promising candidates more effectively than a conventional baseline, while explaining each score?

3. Data and labels

We used NASA's TOI catalog (8,148 rows, 91 columns), a table of catalog values rather than raw light curves.

Disposition	Count	Treated as
CP, KP (confirmed / known planet)	1,420	real (1)
FP, FA (false positive /	1,414	fake (0)

Disposition	Count	Treated as
false alarm)		
PC, APC (undecided)	5,312	to be ranked
No verdict	2	discarded

We split the 2,834 labeled rows 80/20 into a study set (2,262) and an exam set (572), keeping all candidates from the same star on the same side. Eleven clues were used: orbital period, transit duration, transit depth, TESS magnitude, star temperature, surface gravity, radius and distance, plus three calculated planet columns (planet radius, insolation, equilibrium temperature).

4. Methods

- **Baseline:** rank by transit depth (and, separately, depth divided by its uncertainty).
- **Tougher baseline:** logistic regression.
- **ML model:** XGBoost (300 trees, depth 4, learning rate 0.05), evaluated with ROC-AUC on the exam set and with grouped 5-fold cross-validation.
- **Clue groups (ablation):** transit only (4 clues), transit + star (8), all clues (11).
- **Explanation:** SHAP on the transit + star model, chosen because the three calculated planet columns are derived from other columns and would make explanations circular.
- **Bias tests:** a transit-only model, and a distance-matched model trained with equal numbers of real and fake candidates in each of 10 distance groups.
- **Real-data check:** TESS 2-minute light curves via lightkurve, flattened and folded, with signal-to-noise, odd/even depth and secondary-dip tests.

5. Results

5.1 ML versus the baseline (exam-set AUC)

Method	Transit only	Transit + star	All clues
Depth baseline	0.552		
Depth / uncertainty baseline	0.496		
Logistic regression	0.648	0.735	0.770
XGBoost	0.759	0.898	0.924

Cross-validation on the study set averaged 0.912 (folds 0.904 to 0.934), consistent with the exam score. Transit clues alone beat the baseline by about 0.21 AUC. The largest jump came from adding star properties (0.759 to 0.898). All figures come from a single train/test split, so differences of about 0.01 should not be treated as real.

5.2 Explanations and selection bias

SHAP showed physically sensible effects: very short periods push the score down and deeper dips push it up. It also showed large effects from star distance, temperature and size. In the labeled data the fraction of real planets falls steadily with distance:

Distance group	Median distance	Fraction real
Closest	36 pc	0.86
Middle	235 pc	0.43
Farthest	1,019 pc	0.21

Part of this is probably astronomers following up nearby stars first, and part may be genuinely cleaner data. We cannot separate the two with this table.

Model	Full exam AUC	Distance-balanced exam AUC
Star-aware	0.898	0.870
Distance-matched	0.865	0.875

The star-aware model lost only 0.028 on the balanced exam, so most of its skill is not the distance shortcut, and the two models tie within noise there. But the rankings differ strongly:

Ranking of undecided candidates	Median distance of top 100
Star-aware	62 pc
Transit-only	145 pc
Distance-matched	594 pc
All undecided	455 pc

Only 25 of 100 candidates were shared between the star-aware and transit-only top-100 lists (rank similarity 0.54), and 21 of 100 between the distance-matched and star-aware lists. The distance-matched ranking appears to overshoot toward distant stars with deep dips; this is a hypothesis we did not test. We therefore treat candidates appearing in several rankings as the most robust, and none of the three rankings as ground truth.

5.3 Light-curve vetting of 30 top candidates

We selected 30 bright (TESS magnitude 11 or brighter), deep-dip candidates that ranked high in all three rankings (worst rank 269 to 750 of 5,312, roughly the top 5% to 14%), and downloaded up to 6 sectors of 2-minute data each. A flag counted only if it was both significant (over 3 sigma) and large (over 20% of the dip depth), with at least 50 data points per test.

Outcome	Count
Dip seen, passes clearly	13
Dip seen, marginal signal-to-noise (7 to 10)	4
Dip seen, small flag only (TOI 5287.01,	4

Outcome	Count
3460.01, 1124.01, 3344.01)	
Dip seen, binary checks not possible (TOI 1894.01, 6075.01, 6873.01)	3
Inconclusive (too shallow to test)	3
No 2-minute data	3

All 24 testable candidates showed a dip and none showed a large eclipsing-binary signature. This is a sanity check, not validation: the catalog was built from similar TESS data, so seeing the dip again is partly expected; the sample is bright, deep-dip and not random; and “passes” is not “confirmed planet”. TOI 1124.01 (odd/even difference 16 sigma, about 8% of depth) is the best candidate for a closer look with more data.

5.4 Light-curve features pilot

For 275 labeled candidates with 2-minute data (first 2 sectors each), we measured four light-curve clues: signal-to-noise, measured-to-catalog depth ratio, odd/even difference and secondary-dip depth. Grouped 5x5 cross-validation gave:

Clue set	AUC
Transit only	0.675
Transit only + light curve	0.765
All catalog	0.857
All catalog + light curve	0.884

On the transit-only baseline, depth ratio alone gained +0.073 of the +0.089 total, while the eclipsing-binary pair (odd/even and secondary) gained about +0.045. Adding the four star columns instead gained +0.181. The light-curve clues were only weakly correlated with stellar properties (absolute correlation at most 0.24). On the all-catalog baseline the gains were only about +0.01 to +0.02 per clue, and the physics pair gained no more than the signal-to-noise and depth-ratio pair (+0.024 each).

We could not determine whether the depth-ratio effect reflects real physics (fake signals failing to reproduce at the catalog depth) or a bias (better-studied candidates have more accurate catalog timings, giving cleaner folds). Odd/even difference was strongly correlated with signal-to-noise (-0.70), so it is partly a noise measure.

6. Corrections made along the way

Several early conclusions were revised after checking, which we regard as part of the result:

1. **The 0.924 headline was partly circular.** The top clues were calculated planet columns, so we moved to the transit + star model for explanations and reported all clue groups.

2. **TOI 3344.01 was first labeled "dip not found".** A BLS period search found a clean dip of about 7,500 ppm at a period only 4 minutes from the catalog value. Over 757 days of data, that small error smeared the fold. Catalog timing drift can hide real dips.
3. **TOI 6873.01 was first labeled a likely eclipsing binary** from a 7.3-sigma secondary dip. The secondary window held only 36 data points, so the flag was a sparse-data artifact. It is now "binary checks not possible".
4. **"Weak signal" was a misleading label.** It was split into "too shallow to test" and "dip not found at catalog timing".
5. **Distance matching overshot.** It removed the nearby-star preference but pushed the top 100 farther than the typical candidate.

7. Limitations

1. **Selection bias:** labels reflect which candidates astronomers chose to follow up, and the model partly learned this.
2. **Distribution shift:** the undecided candidates are farther and fainter than the labeled ones the model learned from.
3. **No ground truth for the ranking:** AUC measures performance on already-labeled candidates only.
4. **Catalog features only** in the main model; no raw light curves.
5. **One train/test split;** the pilot's reshuffles reuse the same rows and are not independent.
6. **Calculated columns** (planet radius, insolation, equilibrium temperature) are derived from other columns.
7. **Small, non-random vetting sample:** bright stars, deep dips, 2-minute data only. Of 25 pilot candidates without 2-minute data, 24 were false positives, so missing data is itself linked to the label.
8. **Scores are not probabilities.** Use them to order candidates.

8. Next steps

- Compare the frozen prediction snapshot (saved 2026-10-10) with NASA's verdicts once more candidates are classified, to see which of the three rankings predicted best.
- Refit timing with BLS for every pilot candidate and test whether the depth-ratio gain survives.
- Vet more candidates with more sectors, including lower-ranked ones, so pass rates can be compared across ranks.
- Add light-curve features for all candidates with 2-minute data, only if the pilot effect is confirmed.
- Compare against a published vetting tool.

9. Reproducibility

All steps are in the accompanying notebook (ExoHunter_clean.ipynb). Data, models, plots and per-candidate results are saved to a Google Drive folder, and the slow sections save after every star, so a Colab disconnect costs at most one candidate. Random seeds are fixed (42).